Local gradient optimization of modular entangling sequences

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(Received 1 Mar 2018; published 30 Apr 2018)

Implementation of logical entangling gates is a component step toward realizing a quantum computer. We use a gradient search optimization approach to find single-gate sequences which, with the addition of a single additional gate, is universally capable of universal logical entangling operations. The resulting sequences are generally entangling subsets of $SU(4)$ gates. The usefulness of the approach allows for application to any two-qubit system regardless of the Hamiltonian details or the requirements of the experimental implementation. This approach is robust to both operation and local gate errors. We also show how the ability of the local gates to entangle is tied to the fidelity of the local operations and the nonlocal strength.

DOI: [illegible]

I. INTRODUCTION

Quantum computing requires both local single-qubit and nonlocal entangling gate operations. Even without quantum computing, quantum dynamics of systems on the logical qubit scale can be performed with an error less than some "quantum error correction threshold," the precise value depending on the particular choice of encoding scheme. Surface codes offer an extremely high threshold of roughly 1% [1], but even then it is desirable to reduce the physical error rate having to do in order to reduce the overhead associated with error correction.

Every generally non-local dynamical operation of a quantum state of the logical subspace of a multiqubit system, from quantum error correction, to [illegible], to simulation of the dynamics in the system parameters in the Hamiltonian. Here we will focus on [illegible] quantum computation on a universal quantum computer. Universality generally depends primarily on the specific gate implementation of the hardware, and can be demonstrated only by showing that a given set of gates can approximate any unitary operation to arbitrary accuracy. In practice, however, the [illegible] operations are often limited by the precision of the control.

In the past we have a method which can produce the effects of general pulse shapes with the appearance of composite pulse sequences. We use a gradient-based optimization method to find a class of single-qubit operations to be combined between several applications of a slightly entangling gate such that the entire sequence performs a perfect entangling gate while suppressing all possible logical errors. The usefulness of our approach allows for application to any system regardless of the system Hamiltonian or interaction of the user with the control. In addition, it performs well not only for [illegible] errors, but also for time-dependent errors with a [illegible] noise.

II. MODEL AND OPTIMIZATION METHOD

In order to examine the error in the numerical simulations we sample an ensemble of noisy system Hamiltonians. This is, as a result of separate error calculations and requires that the optimized single-qubit operations [illegible] are well [illegible] over multiple realizations [9].

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